ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Main Flow Services and Technologies Pvt. Ltd. for providing me with the opportunity to complete this Cybersecurity Internship. This project enhanced my practical knowledge of network security, vulnerability assessment, and penetration testing techniques using industry-standard tools such as Nmap and OpenVAS.

I also extend my thanks to my mentors and everyone who supported me throughout this internship. Their guidance helped me understand the process of identifying security vulnerabilities, analyzing risks, and recommending appropriate remediation measures. This internship significantly improved my technical skills and practical understanding of cybersecurity concepts.

ABSTRACT

This project focuses on performing a Network Security and Vulnerability Assessment using two widely used cybersecurity tools: Nmap and OpenVAS. The assessment was conducted in a safe virtual environment using Kali Linux, Metasploitable2, and VirtualBox.

In the first phase, Nmap was used to discover the target host, identify open ports, detect running services, and gather system information. In the second phase, OpenVAS was used to perform an in-depth vulnerability assessment on the target system. The scan identified several Critical, High, Medium, and Low severity vulnerabilities, including outdated software, default credentials, remote code execution vulnerabilities, insecure services, and weak authentication mechanisms.

The findings were analyzed to understand their security impact, and appropriate mitigation measures were recommended to reduce the associated risks. This project demonstrates the practical application of vulnerability assessment techniques and highlights the importance of identifying and addressing security weaknesses to improve the overall security posture of computer systems.

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INTRODUCTION

Network security is an essential part of protecting computer systems, networks, and sensitive information from cyber threats. Organizations use vulnerability assessment techniques to identify weaknesses in their systems before attackers can exploit them. Regular security assessments help improve system security by detecting outdated software, insecure configurations, weak passwords, and vulnerable network services.

This project focuses on performing a Network Security and Vulnerability Assessment using Nmap and OpenVAS in a controlled virtual environment. The assessment was carried out on the Metasploitable2 virtual machine, a deliberately vulnerable operating system designed for cybersecurity learning and testing. The target system was hosted in VirtualBox, while Kali Linux was used as the attacking machine.

Initially, Nmap was used to discover the target host, identify open ports, detect running services, and gather information about the operating system. After reconnaissance, OpenVAS was used to perform a comprehensive vulnerability assessment. The scan identified multiple Critical, High, Medium, and Low severity vulnerabilities. Based on these findings, suitable mitigation recommendations were prepared to reduce security risks and improve the overall security posture of the target system.

OBJECTIVES

- To understand the fundamentals of network security and vulnerability assessment.
- To perform network reconnaissance using Nmap.
- To identify open ports and running services on the target system.
- To conduct vulnerability scanning using OpenVAS.
- To analyze the identified vulnerabilities based on their severity.
- To understand the security risks associated with vulnerable services.
- To recommend appropriate mitigation techniques for reducing security risks.
- To gain practical experience with cybersecurity tools in a virtual environment.

TOOLS USED

S. No	Tool	Version	Purpose	Platform
1	Kali Linux	Latest	Attacking machine used for security testing	Virtual Machine
2	Nmap	Latest	Network scanning and reconnaissance	Kali Linux
3	OpenVAS (Greenbone)	Latest	Vulnerability assessment and scanning	Kali Linux
4	Metasploitable2	Ubuntu 8.04	Vulnerable target machine	Virtual Machine
5	Oracle Virtual Box	Latest	Virtualization platform for running VMs	Windows 11

METHODOLOGY

1. Network Setup

A virtual cybersecurity testing environment was created using Oracle VirtualBox. Kali Linux was configured as the attacker machine, while Metasploitable2 served as the vulnerable target machine. Both virtual machines were connected through the same Host-Only Network, enabling secure communication without affecting external systems.

2. Network Reconnaissance using Nmap

The target IP address (192.168.56.102) was identified, and Nmap was used to perform network reconnaissance. The scan detected active hosts, open ports, running services, and operating system details. This information helped identify potential entry points for further security analysis.

3. Vulnerability Assessment using OpenVAS

The target system was scanned using OpenVAS (Greenbone Vulnerability Manager). After updating the vulnerability database, a full vulnerability scan was executed. OpenVAS analyzed the target system and identified Critical, High, Medium, and Low severity vulnerabilities based on known security issues.

4. Analysis and Recommendations

The scan results were analyzed to understand the security risks associated with each vulnerability. Appropriate mitigation techniques and security recommendations were prepared to reduce the identified risks and improve the overall security posture of the target system.

Nmap NETWORK RECONNAISSANCE

The first phase of the project involved network reconnaissance using Nmap. The objective was to identify the target machine, discover open ports, detect running services, and gather system information. The target machine used for scanning was Metasploitable2 with the IP address 192.168.56.102. The reconnaissance results provided valuable information for the vulnerability assessment performed later using OpenVAS.

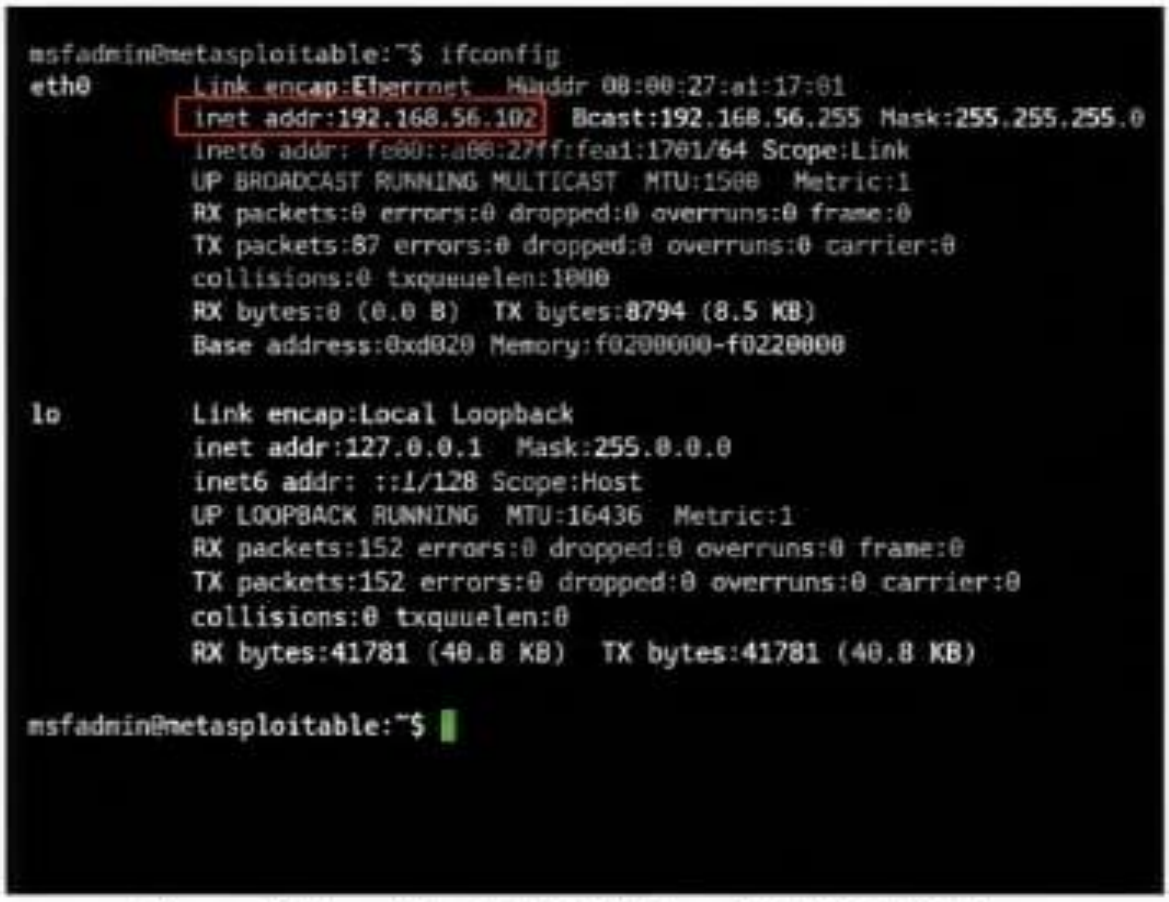


Figure 1: Metasploitable 2 IP Address (192.168.56.102)

Updating Kali Linux Packages

Before performing security testing, the Kali Linux system was updated using the sudo apt update command. Updating the package repository ensures that all installed tools have the latest security patches and package information, improving stability and compatibility during the assessment.

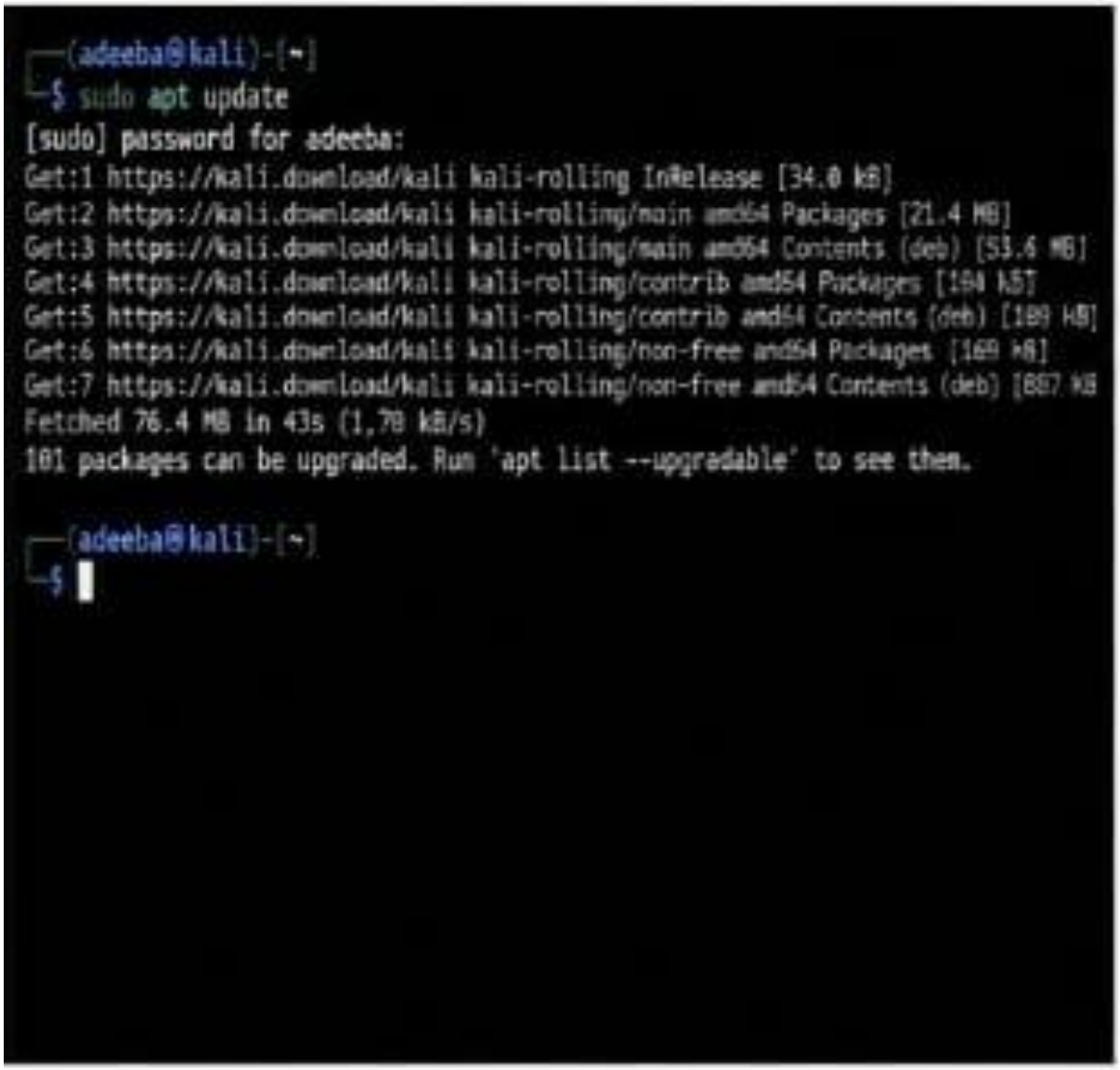


Figure 2: Updating Kali Linux Packages

GVM (OpenVAS) Setup Verification

The GVM (Greenbone Vulnerability Management) installation was verified using the gvm-check-setup command. This verification confirmed that the required services, certificates, databases, and vulnerability feeds were properly configured and ready for vulnerability scanning.



Figure 3: GVM (OpenVAS) Setup Verification

Starting Greenbone Vulnerability Manager

The Greenbone Vulnerability Manager (GVMD) service was started to initialize the OpenVAS framework. This step prepares the web interface and scanning services required for vulnerability assessment.



Figure 4: Starting Greenbone Vulnerability Manager

Disk Space Verification

The available storage space and partition details were verified using the df -h and lsblk commands. This ensured that sufficient disk space was available for installing updates, vulnerability feeds, and scan reports.

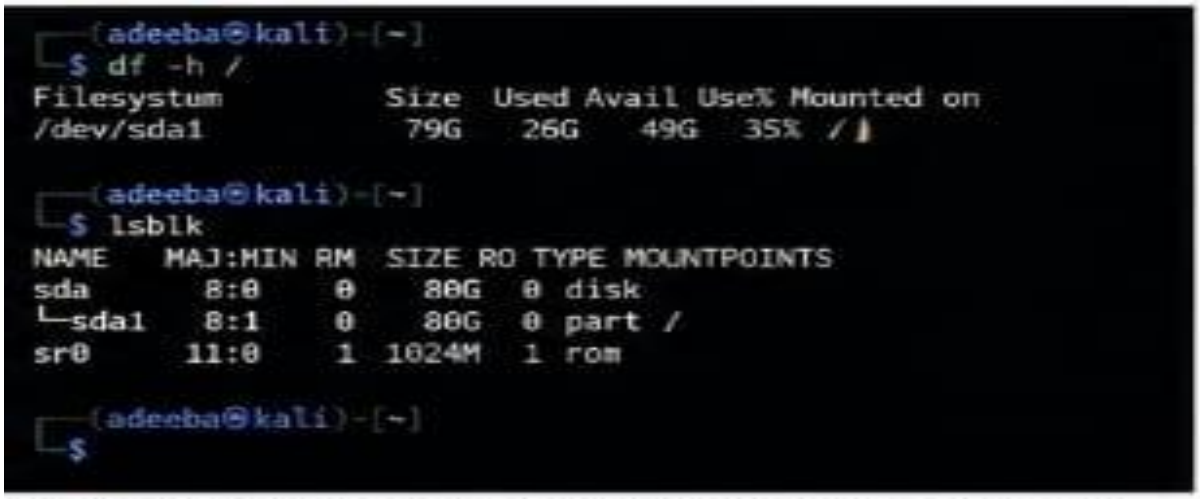


Figure 5: Disk Space Verification (79G Total, 49G Available)

PostgreSQL Service Status

The PostgreSQL database service was restarted and verified because OpenVAS depends on PostgreSQL for storing vulnerability data and scan results. The service status confirmed that the database was active and functioning correctly.



Figure 6: PostgreSQL Service Status (Active)

Nmap Scan Command

Network reconnaissance was performed using the Nmap utility. The command scanned the target IP address to identify active hosts, open ports, and running services before conducting vulnerability analysis.



Figure 7: Nmap Scan Command

Nmap Scan Results

The Nmap scan successfully identified multiple open ports and network services running on the target machine. These results provided the information required to perform further vulnerability assessment.



Figure 8: Nmap Scan Results (Top Open Ports)

Service Version Detection

Nmap service version detection was performed to identify the software versions running on each open port. This information helps determine potential vulnerabilities associated with outdated or insecure services.



Figure 9: Nmap Service Version Detection

VULNERABILITY ANALYSIS

The results obtained from Nmap and OpenVAS were analyzed to identify security weaknesses in the Metasploitable2 target machine. Multiple open ports and outdated services were detected, increasing the risk of unauthorized access. OpenVAS categorized the identified vulnerabilities based on severity levels such as Critical, High, Medium, and Low. These findings highlight the importance of timely patching, secure configurations, and regular vulnerability assessments to improve system security.

RECOMMENDATIONS

- Regularly update the operating system and installed software.
- Close unnecessary open ports and disable unused services.
- Use strong authentication and secure passwords.
- Enable firewall protection and restrict unauthorized access.
- Perform periodic vulnerability scanning using OpenVAS.
- Monitor systems continuously for suspicious activities.

CONCLUSION

This project successfully demonstrated the process of network reconnaissance and vulnerability assessment using Kali Linux, Nmap, OpenVAS, and Metasploitable2. The identified vulnerabilities and open services highlighted potential security risks in the target system. The project also emphasized the importance of regular security assessments and implementing appropriate mitigation techniques to strengthen cybersecurity.

REFERENCES

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https://www.kali.org/docs/

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Greenbone/OpenVAS Documentation –

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https://sourceforge.net/projects/metasploitable/